Panskura Banamali College (Autonomous)

B.Sc.(H) 2nd Semester Examination -2021 (Online Mode)

Sub: Computer Science **Paper: CC4T**

Paper Name: Discrete Structures

F.M.: 60 **Time : 3 Hours**

Answer any four : **4x15=60**

1. i. Differentiate between the difference and symmetric difference of two sets. For three subsets A, B , C of universal set S , prove that $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$.

ii. If R and S be two equivalence relations on a set A , then prove or disprove that $R \cap S$ is an equivalence relation the set A. What is an equivalence class ? Find the equivalence class for the relation $R = \{ (x,y) : x,y \varepsilon z$ and x-y is divisible by 4 $\}$ on z , if exist.

iii. If $f : A \rightarrow B$ and $g: B \rightarrow C$ be two mappings such that $g \circ f : A \rightarrow C$ is injective , then prove that f is injective . (2+4)+(3+1+2)+3

2. i. Define n – any relation on a set. Give an example of a binary relation which is reflexive and transitive but neither symmetric nor asymmetric . How many reflexive relations are there on a set of n elements ? Evaluate .

ii. What is meant by partition of a set ? Find a partition of the set AUB where A and B are nonempty sets.

iii. Find a formula for 1/1.2 + 1/2.3+---- + 1/n.(n+1) by examining the values of this expression for small values of n . Prove this formula using mathematical induction. (1+3+3) +(2+2) + 4

3. i. Suppose that $f_1(x)$ is $O(g_1(x))$ and that $f_2(x)$ is $O(g_2(x))$, then prove that $(f_1+f_2)(x)$ is $O(\max(|g_1(x)|,|g_2(x)|))$. Give a big – O estimate for $f(n) = 3n \log(n!) + (n^2+3) \log n$.

ii. State the principle of inclusion and exclusion.

iii. Show that if n and k are positive integers and $n > k$, then $\binom{n+1}{k} = (n+1)\binom{n}{k-1}/k$. Find the co-efficient of $x^{101} y^{99}$ in the expression of $(7x-3y)^{200}$. (4+3)+2+(4+2)

4. i. State master theorem. Find a recurrence relation and give initial condition for the number of bit strings of length n that do not condition two consecutive 0' s. Is this relation a linear relation ? How many such bit strings are of length 7 ?

ii. Find the generating function for the sequence { 1,3,9,27 -------- } . Using generating function solve the recurrence relation $a_k = 4a_{k-1} - 4a_{k-2}+k^2$, $a_0=2$,$a_1=5$. (2+4+1+2)+ (1+5)

5. i. Define graph isomorphism. Show that the maximum degree of any vertex in a simple graph with n vertices is (n-1) .

ii. Prove that a simple graph with n vertices and k components can have at most $\frac{(n-k)(n-k+1)/}{2}$ edges.

iii. Define logical equivalence with suitable example. hence show that $(p \rightarrow q) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$.

(1+4)+ 5+ (1+4)

6. i. Show that $\sim[p \wedge q) \rightarrow (p \vee q)]$ is a contradiction.

ii. Define tree .Prove that any connected graph with n vertices and (n-1) edges is a tree.

iii. Prove that a connected graph is an Euler graph if and only if all vertices of this graph are of even degree.

5+5+5

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